SAP BRIM: Deep Dive into the Four Modules

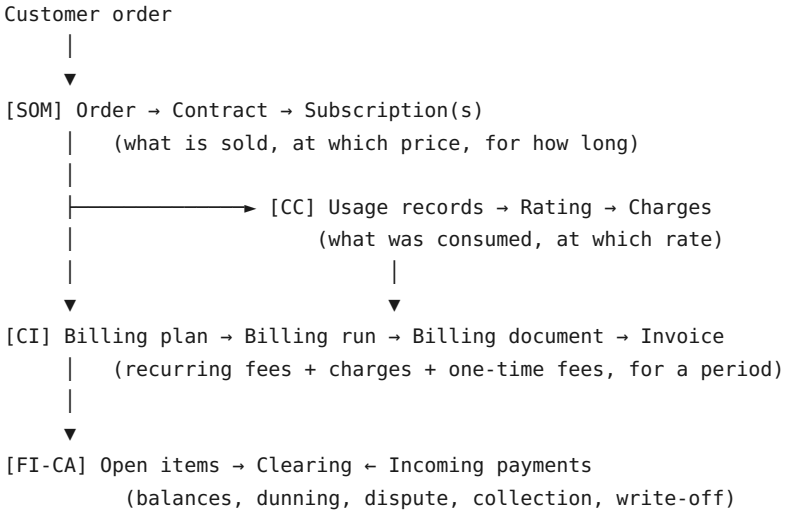
Purpose: Component-level reference for the four core modules of SAP Billing and Revenue Innovation Management (BRIM). Each component is explained in terms of what it is, what it does, how it connects to other components, and what to check when it misbehaves.

Modules covered 1. Subscription Order Management (SOM) 2. Convergent Charging (CC) 3. Convergent Invoicing (CI) 4. Contract Accounts Receivable and Payable (FI-CA)

Important: Object names, transaction codes, Fiori apps, and OData services differ by SAP release (S/4HANA on-premise, Cloud Public Edition, or older CRM-based setups). Terms below describe the business and functional meaning. Verify exact names in the client's system before using them in answers.

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0. How the four modules fit together



Key principle: SOM decides *what* is owed and *from when*. CC decides *how much* usage is worth. CI decides *when* and *how* to bill it. FI-CA decides *how the money is tracked and collected*.

Key identifiers that link the modules - Business partner: the customer, shared across all modules - Contract account: the financial account in FI-CA that receives billing documents - Contract: the commercial agreement in SOM, linked to a contract account - Subscription: the product instance inside a contract, which drives charges - Usage record: linked to a subscription or contract through its identifiers

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Module 1: Subscription Order Management (SOM)

Role in BRIM: The front end of the commercial lifecycle. SOM captures what the customer buys and manages changes over time. Everything billed later originates here.

1.1 Product catalog and product model

What it is: The set of products and services the company sells, with their structure and attributes.

Components: - **Product:** A sellable item, such as a mobile plan, an internet package, or a software subscription. - **Product attributes:** Characteristics that vary, such as speed tier, data allowance, or contract term. - **Product hierarchy:** Bundles and components. A bundle contains several products. Each component can have its own pricing. - **Product validity:** Start and end dates that control when a product can be sold.

What to check when it fails: - Product not available for selection: validity dates, sales organization, or channel assignment - Wrong attributes shown: product configuration, not contract data

1.2 Product proposal and offers

What it is: The process of presenting a product or bundle to a customer before an order is placed.

Components: - **Offer or proposal:** A quotation-like object that holds the proposed products, prices, and terms. - **Conversion:** Turning an accepted proposal into an order.

Why it matters for a chatbot: Questions like “What plans are available?” come from the product catalog and offers, not from contracts.

1.3 Price elements and price determination

What it is: The mechanism that calculates what the customer pays for a product.

Components: - **Price element:** A type of price, for example recurring fee, one-time fee (setup fee), or usage-based fee. - **Price determination:** The steps that select and calculate the prices for an order item, based on product, customer, validity, and conditions. - **Discounts and surcharges:** Adjustments applied on top of the base price. These can be percentage-based or amount-based. - **Price validity:** Prices can change over time. The price valid on the order date or subscription start date is the one used.

What to check when it fails: - Unexpected price: check the price element assignment and any discount conditions - Price not found: check that the price is maintained for the product, currency, and validity date

1.4 Subscription order

What it is: A request from a customer (or from an agent on their behalf) to create, change, or cancel products.

Components: - **Order header:** Customer, sales channel, order date, and requested start date - **Order items:** Each product in the order, with its action - **Item action:** Create, change, suspend, resume, or cancel - **Order status:** Open, in process, completed, or rejected (confirm status names for the release) - **Requested date:** The date the change should take effect

Typical flow: 1. Customer selects products (from an offer or directly) 2. Order is created with items and prices 3. Order is validated (for example credit check or eligibility) 4. Order is completed, which creates or changes the contract and subscriptions

What to check when it fails: - Order stuck or rejected: validation rules, missing data such as contract account or address, or credit or eligibility checks - Order completed but wrong date: requested date and pro-rating rules

1.5 Contract

What it is: The commercial agreement with the customer. It holds the products, terms, and link to the contract account.

Components: - **Contract header:** Business partner, contract account, contract start date, and contract end date (if defined) - **Contract items or subscriptions:** The products in force - **Contract terms:** Payment terms, billing terms, and cancellation terms (which vary by product and business rules) - **Contract status:** Active, terminated, or other statuses defined in configuration

Relationship to billing: The contract tells billing which contract account to bill, which products to bill, and on which billing plan.

1.6 Subscription

What it is: A product instance inside a contract with its own start date, end date, and status. A subscription is what drives recurring billing.

Components: - **Subscription ID:** Unique identifier used in billing and usage matching - **Start and end dates:** Control when recurring fees begin and end - **Subscription status:** Lifecycle state, for example active, suspended, or terminated - **Product reference:** Which product and price the subscription uses - **Billing relevance:** Whether the subscription generates recurring charges

Subscription lifecycle:

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Created → Active → (Suspended ↔ Active) → Terminated/Cancelled
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What to check when it fails: - Subscription billed when it should not be: check status and end date - Subscription not billed: check status, billing relevance, and billing plan link

1.7 Changes: upgrade, downgrade, and cancellation

What it is: Modifications to an existing contract or subscription.

Components: - **Change order:** An order that modifies existing subscriptions - **Upgrade:** A new or higher-priced product replaces or is added to an existing one - **Downgrade:** A lower-priced product replaces an existing one - **Cancellation:** Ends a subscription from a given date. Cancellation rules (notice period, minimum term, fees) are defined by the client's policy and configuration. - **Pro-rating:** How partial periods are charged when a change happens mid-period. Pro-rating rules are configured per product and billing period.

What to check when it fails: - Wrong credit or charge after a change: pro-rating rules and effective date - Cancellation not effective: effective date or notice period

1.8 Usage-based products in SOM

What it is: Products whose charges depend on consumption. SOM defines them, but CC calculates the amount.

Components: - **Usage type link:** Connects the product to the usage types that rating will use - **Rate plan link:** Connects the product to the pricing used in rating - **Usage allowance:** Included quantities (for example 10 GB included) before extra usage is charged

1.9 SOM integration points

- **Business partner and CRM:** Customer data for orders
- **S/4HANA Sales and Service:** Order and quotation integration in some setups
- **Convergent Charging:** Passes product and rate information for usage
- **Convergent Invoicing:** Provides subscription data that drives recurring billing

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Module 2: Convergent Charging (CC)

Role in BRIM: Takes usage (events, records, meter readings) and turns it into charges with monetary values. It also handles the rules that decide which price applies to each usage record.

2.1 Usage records (event data)

What it is: The raw data describing consumption. Examples: a phone call, a data session, an SMS, an electricity meter reading, or a software license use.

Components: - **Event or usage record:** One unit of consumption, with attributes such as time, quantity, and identifiers - **Usage type:** The category of consumption (for example voice, data, SMS) - **Attributes:** Details used for pricing, such as zone, time band, service type, or roaming status - **Identifiers:** Links to the subscription or contract, such as a subscription ID or telephone number

Sources: Usage records usually come from a network, a metering system, or an application. They are normally delivered as files or messages.

2.2 Mediation (conceptual)

What it is: The step that receives raw usage from source systems, checks it, normalizes it, and passes it to rating. In many SAP landscapes, mediation is handled by a separate product (SAP Convergent Mediation, or a third-party system).

Typical tasks: - Collect and validate incoming files or messages - Remove duplicates - Map raw fields to the standard attributes BRIM expects - Route records that fail validation to an error queue

What to check when it fails: - Usage missing from billing: check whether it reached mediation and whether it failed validation - Duplicate charges: check duplicate detection

2.3 Rate plans and rate determination

What it is: Rate plans define the prices for usage. Rate determination selects the correct rate for each usage record.

Components: - **Rate plan:** A set of rates for a product or group of products - **Rate table or rate elements:** The actual prices, tiered or flat, by usage type and attributes - **Rate determination logic:** The rules that match a usage record to a

rate using its attributes, such as usage type, zone, time band, and validity date - **Tiers and bands:** Ranges where the rate changes, for example the first 100 units at one price and the rest at another

What to check when it fails: - Usage not rated: no rate found for the attributes. Check usage type, zone, time band, and validity dates. - Wrong rate applied: check rate determination order and attribute mapping - Rate changed unexpectedly: check the validity date of the rate plan

2.4 Charge types and charge creation

What it is: A charge is the monetary result of rating. Charge types categorize charges so billing and reporting can handle them correctly.

Components: - **Charge type:** Category of charge, such as usage, recurring fee, one-time fee, discount, or tax-related charge - **Charge item:** A single calculated charge with quantity, amount, currency, and reference to the usage or subscription - **Charge creation:** The step that stores the charge so billing can pick it up

2.5 Rating

What it is: The calculation that turns a usage record into a charge. It combines quantity, rate, and any rules for rounding, tiers, and discounts.

Components: - **Rating step:** Applies the selected rate to the quantity - **Rounding rules:** Determine how amounts are rounded (confirm with the finance team) - **Tier logic:** Applies different rates to different quantity ranges - **Discounts at rating level:** Some discounts are applied to usage before billing

2.6 Real-time and batch rating

Batch rating: Usage is collected and rated in scheduled runs. Used for most postpaid usage.

Real-time rating: Usage is rated as it arrives. Used for prepaid or balance-controlled services, where a decision must be made during the event.

Trade-off: Real-time is faster and needed for credit control, but it is more complex to run and monitor. Batch is simpler and common for billing-only use cases.

2.7 Discounts, credits, and allowances

Components: - **Discount rule:** Reduces a charge based on conditions such as customer segment or volume - **Credit:** A negative charge, for example for service issues - **Allowance or free units:** Included quantity that is not charged until used up. Tracking is done per period.

What to check when it fails: - Allowance not applied: check period boundaries and the subscription that holds the allowance - Discount missing: check conditions and whether the discount is valid on the usage date

2.8 Error handling and suspense

What it is: Records that cannot be rated or matched are not discarded. They are held for review.

Components: - **Error or suspense queue:** Holds records that failed rating or matching - **Reason codes:** Explain why a record failed - **Reprocessing:** After the cause is fixed, records are rated again

Operational importance: A growing error queue is a common cause of incorrect or late invoices. Monitor it.

2.9 Rerating

What it is: Recalculating charges that were already rated, usually after a rate correction or a late change.

Impact: Rerating can change charges already sent to billing. Plan how existing billing documents will be corrected, usually through credit and new invoices (policy decision).

2.10 CC integration points

- **SOM:** Provides subscription and product links for matching usage
- **CI:** Receives charges to bill
- **Mediation and usage sources:** Provide raw records

Module 3: Convergent Invoicing (CI)

Role in BRIM: Collects everything that should be billed for a contract account in a period (recurring fees, one-time fees, and charges from rating), calculates taxes, and creates the billing documents and invoices.

3.1 Contract account in billing

What it is: The billing account that receives the billing documents. CI bills at the contract account level.

Components: - **Contract account link:** Each billing document belongs to one contract account - **Payment and billing terms:** Due dates, payment methods, and billing cycles assigned to the account - **Billing relevant data:** Data that influences how billing is done for the account

3.2 Billing plan

What it is: Defines when and how billing happens for a contract. It is the schedule and the rules for billing periods.

Components: - **Billing cycle:** The regular period, such as monthly or quarterly - **Billing date:** The day the billing run treats as the billing date - **Billing periods:** The start and end of each period billed - **Due date rules:** Calculation of when payment is due, usually based on the invoice date - **Anchor dates:** Dates used to align the billing cycle, such as a contract start date

What to check when it fails: - Billing on the wrong date: check the billing plan's billing date and anchor - Period gaps or overlaps: check the contract start date and any billing plan changes

3.3 Billing run

What it is: A batch process that creates billing documents for many contract accounts at once.

Components: - **Selection:** Which contract accounts or billing plans are included - **Run parameters:** Billing date, billing period, and mode (test or live) - **Logs:** Results for each account, including errors and warnings - **Restart:** Failed accounts can be rerun after the issue is fixed

Common operational practice: - Run billing simulation before the live run - Schedule runs away from peak usage times - Check the error log before releasing invoices

3.4 Billing simulation

What it is: A test run that shows what billing would produce without posting the results.

Use cases: - Explain expected charges before a live run - Check a new product or rate before it is used - Compare results against the previous period

3.5 Billing document

What it is: The internal record of the billing for a contract account and billing period. It contains the lines and totals.

Components: - **Document header:** Contract account, billing date, billing period, and document number - **Line items:** Each charge, fee, discount, or tax line - **Tax information:** Tax codes and amounts per line - **Totals:** Net amount, tax, and gross amount - **Status:** Created, released, or cancelled (confirm names for the release)

Why it matters: Billing documents are the source for invoices and the link back to usage and subscriptions. Investigating a billing question usually starts here.

3.6 Invoice

What it is: The customer-facing document created from one or more billing documents. It is sent to the customer in print or electronic form.

Components: - **Invoice header:** Invoice number, invoice date, due date, customer details, and payment information - **Invoice lines:** Formatted from billing document lines, often grouped by product or service - **Totals and tax summary** - **Layout or form:** The print template that defines how the invoice looks

Relationship to FI-CA: When an invoice is released, an open item is created in the contract account in FI-CA.

3.7 Invoice forms and output

Components: - **Print form:** Layout used for printing or PDF output - **Output channel:** Mail, e-mail, or electronic delivery - **Invoice details by line or summary:** Level of detail controlled by configuration

3.8 Taxes

Components: - **Tax determination:** Assigns tax codes to lines based on the customer's location, product type, and rules - **Tax calculation:** Calculates tax amounts - **Tax reporting:** Supports statutory reporting in finance

What to check when it fails: Wrong tax rate or tax code. Check tax determination rules and the customer's tax data.

3.9 Invoice correction and reversal

Reversal: A billing document or invoice is cancelled. The open item is reversed in FI-CA.

Rebilling: A new billing document is created after reversal, with corrected data.

Credit memo: A separate document that reduces the amount owed, often used for corrections or goodwill credits.

Policy point: The correction method (reversal and rebilling, credit memo, or rerating) must follow the client's accounting and customer service policy.

3.10 CI integration points

- **SOM:** Provides subscriptions and recurring fees
- **CC:** Provides usage charges
- **FI-CA:** Receives open items and posts payments
- **General ledger:** Receives revenue and tax postings

Module 4: Contract Accounts Receivable and Payable (FI-CA)

Role in BRIM: Manages the financial side of the customer relationship. It tracks what the customer owes, what they have paid, what is overdue, and how overdue amounts are collected. FI-CA is the subledger for receivables.

4.1 Business partner

What it is: The master record for the customer. It holds identity, address, bank details, and roles.

Components: - **Business partner ID:** Unique identifier - **Roles:** For example customer or contact person - **Addresses and bank details:** Used for correspondence and payments (sensitive data, restricted access) - **Relationships:** Links between business partners, for example a parent and subsidiary

Relationship to FI-CA: A business partner can have one or more contract accounts.

4.2 Contract account

What it is: The financial account where receivables and payables are tracked for a business partner.

Components: - **Contract account ID:** Unique identifier - **Contract account category:** Defines the account type and its rules - **Payment terms and methods:** How the customer pays - **Dunning procedure assignment:** Which dunning rules apply - **Account balance:** The total of open items, split into overdue and not overdue - **Account status:** Active, blocked, or closed

Why it matters: Most chatbot balance and payment questions are answered from the contract account.

4.3 Document types and posting

What it is: Financial documents that record receivables, payments, and adjustments.

Components: - **Document types:** Classify postings, for example invoice, payment, credit memo, or reversal - **Posting keys and account determination:** Decide which accounts are debited or credited - **Document number ranges:** Control numbering

Relationship to billing: Each invoice release posts a receivable document in FI-CA.

4.4 Open items

What it is: Receivables or payables that are not yet settled.

Components: - **Open item:** Amount, due date, document reference, and status - **Overdue indicator:** Whether the due date has passed - **Open item status:** Open, cleared, partially cleared, or reversed

Typical queries: Which items are overdue? What is the total open? Which invoices are still unpaid?

4.5 Clearing

What it is: The matching of payments to open items. Clearing closes items when they are settled.

Types: - **Automatic clearing:** Matches payments by reference (for example invoice number or payment reference) - **Manual clearing:** A user matches items manually - **Partial clearing:** A payment covers only part of an item - **Net clearing:** Receivables and payables for the same account are netted

What to check when it fails: - Payment not matched: missing payment reference or incorrect reference format - Item still open after payment: check clearing and amount

4.6 Payments

What it is: Money received from customers and processed in FI-CA.

Components: - **Payment methods:** Bank transfer, direct debit, card, cash, or other - **Payment lots:** Groups of payments processed together, for example from one bank file - **Incoming payment posting:** Records the payment and its allocation - **Payment allocation rules:** Define which open items receive the money when a payment does not specify items - **Returned payments:** Bank returns for direct debit or failed transfers. The original items are reopened.

Direct debit: The company collects payment from the customer's bank account using a mandate. Mandate management and return handling are important controls.

4.7 Payment plans and installments

What it is: Agreements that split an overdue or large amount into scheduled payments.

Components: - **Installment plan:** Schedule of amounts and dates - **Deferral:** Postponement of due dates - **Monitoring:** Tracking of plan compliance

Policy point: Eligibility, approval, and terms must come from the client's policy.

4.8 Dunning

What it is: The process of reminding customers about overdue items and escalating if they do not pay.

Components: - **Dunning procedure:** A set of rules assigned to contract accounts - **Dunning levels:** Stages such as reminder 1, reminder 2, and final notice - **Dunning run:** The batch process that evaluates accounts and issues notices - **Dunning notices:** Letters or emails sent to the customer - **Dunning blocks:** Stops dunning for an account or item (for example during a dispute) - **Dunning charges:** Fees added at certain levels (policy decision)

What to check when it fails: - Customer received a reminder for a paid item: check clearing and the dunning block - No reminder sent: check the procedure assignment and the due date

4.9 Collections and write-off

Collections: Actions after dunning, such as handing the account to a collection agency or taking legal steps. Requires policy and vendor integration.

Write-off: Removing an uncollectable amount from receivables. This is an accounting decision, usually under finance control and policy.

4.10 Interest

What it is: Charges calculated on overdue amounts, if the client's policy and contracts allow it.

Components: - **Interest rules:** Rate, calculation method, and start date - **Interest run:** Batch process that calculates and posts interest

Policy point: Rates and rules must come from the client's approved terms. Do not state a rate without the source document.

4.11 Dispute management (FI-CA-DM)

What it is: Management of customer disputes about invoices or items.

Components: - **Dispute case:** Record of the dispute, with reason and status - **Disputed items:** Open items linked to the case - **Dunning block:** Disputed items can be blocked from dunning - **Resolution:** Outcome, which may include correction, credit, or rejection

Operational point: Dispute handling must be consistent with the client's customer service policy and deadlines.

4.12 Correspondence and documents

What it is: Letters and notices sent from FI-CA, including dunning notices, account statements, and confirmations.

Components: - **Correspondence type:** Template and trigger - **Output channel:** Print, e-mail, or other - **Statements:** Periodic summaries of balance and activity

4.13 Refunds and credit balances

What it is: When a contract account has a credit balance, the company may refund it to the customer.

Components: - **Credit balance:** Amount the company owes the customer - **Refund process:** Payment run that transfers the money - **Approval and controls:** Follow client policy

Policy point: Refund eligibility, timing, and approval are policy decisions.

4.14 FI-CA integration points

- **CI:** Receives invoices and creates open items
- **Business Partner and CRM:** Customer master data
- **General ledger:** Subledger reconciliation and postings
- **Banks and payment providers:** Incoming and outgoing payments

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5. Cross-module process walkthrough

Scenario: a customer subscribes to a mobile plan, uses data, and pays late.

1. **SOM:** The customer orders a plan. The order is completed, creating a contract with a subscription that starts on the requested date.
2. **CC:** Data usage records arrive from the network. They are rated using the subscription's rate plan. Charges are created for usage above the included allowance.
3. **CI:** The billing plan runs at the end of the month. Recurring fee, usage charges, and taxes are collected into a billing document. An invoice is created.
4. **FI-CA:** The invoice creates an open item on the contract account with a due date.
5. **FI-CA:** The customer does not pay by the due date. The item becomes overdue. The dunning run sends the first reminder.
6. **FI-CA:** The customer pays after the second reminder. The payment is matched to the open item by clearing.
7. **FI-CA:** The contract account balance returns to zero. Dunning stops.

Chatbot relevance: Each step generates a question users may ask. "Why is my bill high?" points to CC and CI. "Has my payment been received?" points to FI-CA. "When does my plan end?" points to SOM.

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6. Configuration overview by module

This is a conceptual checklist, not a configuration guide. Exact steps depend on the release and the client's design.

SOM - Product catalog, product hierarchy, and validity - Price elements and price determination - Order types, item categories, and statuses - Contract and subscription statuses - Pro-rating and cancellation rules

CC - Usage types and attributes - Rate plans and rate tables - Rate determination logic and tiers - Rounding rules - Error queue and reason codes

CI - Billing plans and cycles - Billing run parameters - Tax determination rules - Invoice forms and output channels - Reversal and correction rules

FI-CA - Business partner and contract account categories - Document types and number ranges - Payment methods, lots, and allocation rules - Dunning procedures and levels - Dispute, interest, and write-off rules

7. Common issues by module

Module	Symptom	Likely cause to check first
SOM	Order not completed	Validation error, missing data, credit or eligibility check
SOM	Wrong price on order	Price element assignment, discount conditions, validity dates
SOM	Subscription billed after cancellation	End date, status, billing relevance
CC	Usage not charged	Usage not received, failed validation, no rate found
CC	Charge amount wrong	Rate plan validity, tier logic, rounding
CC	Duplicate charges	Duplicate usage records, mediation duplicate check
CI	Invoice missing a charge	Charge not created, billing cut-off, billing period
CI	Wrong billing date	Billing plan date and anchor
CI	Wrong tax	Tax determination rules, customer tax data
FI-CA	Payment not matched	Reference missing or wrong format
FI-CA	Dunning letter for a paid invoice	Clearing not posted, dunning block missing
FI-CA	Balance does not match invoices	Unposted items, reversal, or clearing error

8. Glossary (extended)

Term	Module	Meaning
Business partner	FI-CA	Master record for a customer
Contract account	FI-CA	Account holding receivables and payables
Product	SOM	Sellable item
Price element	SOM	Type of price, such as recurring or one-time
Order item	SOM	Product line in an order with an action
Contract	SOM	Commercial agreement with the customer
Subscription	SOM	Product instance with dates and status
Pro-rating	SOM	Charging partial periods
Usage record	CC	Raw consumption event
Mediation	CC	Collection and validation of usage before rating
Rate plan	CC	Set of rates for usage
Rating	CC	Calculation of charge from quantity and rate
Charge item	CC	Calculated charge passed to billing
Suspense queue	CC	Holding area for records that failed processing

Billing plan	CI	Schedule and rules for billing a contract
Billing run	CI	Batch creation of billing documents
Billing document	CI	Internal record of billing for a period
Invoice	CI	Customer-facing bill
Open item	FI-CA	Unsettled receivable or payable
Clearing	FI-CA	Matching payments to items
Dunning	FI-CA	Reminder and escalation process
Dispute case	FI-CA	Record of a customer challenge
Write-off	FI-CA	Removal of an uncollectable amount

9. Items to confirm with the client

1. BRIM release and deployment (S/4HANA on-premise, Cloud Public Edition, or other)
2. Whether mediation is a separate product, and which usage sources exist
3. Product catalog, rate plans, and billing cycles in use
4. Dunning levels, charges, and timelines (from approved policy only)
5. Interest, refund, write-off, and cancellation rules (from approved policy only)
6. Tax rules and invoice format requirements
7. Exact object names, transaction codes, and services available in the release
8. Authorization roles for chatbot users

Document control

- **Version:** 1.0 (draft)
- **Owner:** [Client process owner]
- **Prepared by:** [Name]
- **Last reviewed:** [Date]
- **Next review:** Every release upgrade or major policy change